Final Project

Sophie Scherkenbach

2025-04-18

```
rm(list = ls())

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(infer)
library(ggplot2)
library(broom)
library(ISLR)
```

```
glimpse(Credit)
```

```
## Rows: 400
## Columns: 12
## $ ID        <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 1~
## $ Income    <dbl> 14.891, 106.025, 104.593, 148.924, 55.882, 80.180, 20.996, 7~
## $ Limit     <int> 3606, 6645, 7075, 9504, 4897, 8047, 3388, 7114, 3300, 6819, ~
## $ Rating    <int> 283, 483, 514, 681, 357, 569, 259, 512, 266, 491, 589, 138, ~
## $ Cards     <int> 2, 3, 4, 3, 2, 4, 2, 2, 5, 3, 4, 3, 1, 1, 2, 3, 3, 3, 1, 2, ~
## $ Age       <int> 34, 82, 71, 36, 68, 77, 37, 87, 66, 41, 30, 64, 57, 49, 75, ~
## $ Education <int> 11, 15, 11, 11, 16, 10, 12, 9, 13, 19, 14, 16, 7, 9, 13, 15,~
## $ Gender    <fct>  Male, Female,  Male, Female,  Male,  Male, Female,  Male, F~
## $ Student   <fct> No, Yes, No, No, No, No, No, No, No, Yes, No, No, No, No, No~
## $ Married   <fct> Yes, Yes, No, No, Yes, No, No, No, No, Yes, Yes, No, Yes, Ye~
## $ Ethnicity <fct> Caucasian, Asian, Asian, Asian, Caucasian, Caucasian, Africa~
## $ Balance   <int> 333, 903, 580, 964, 331, 1151, 203, 872, 279, 1350, 1407, 0,~
```

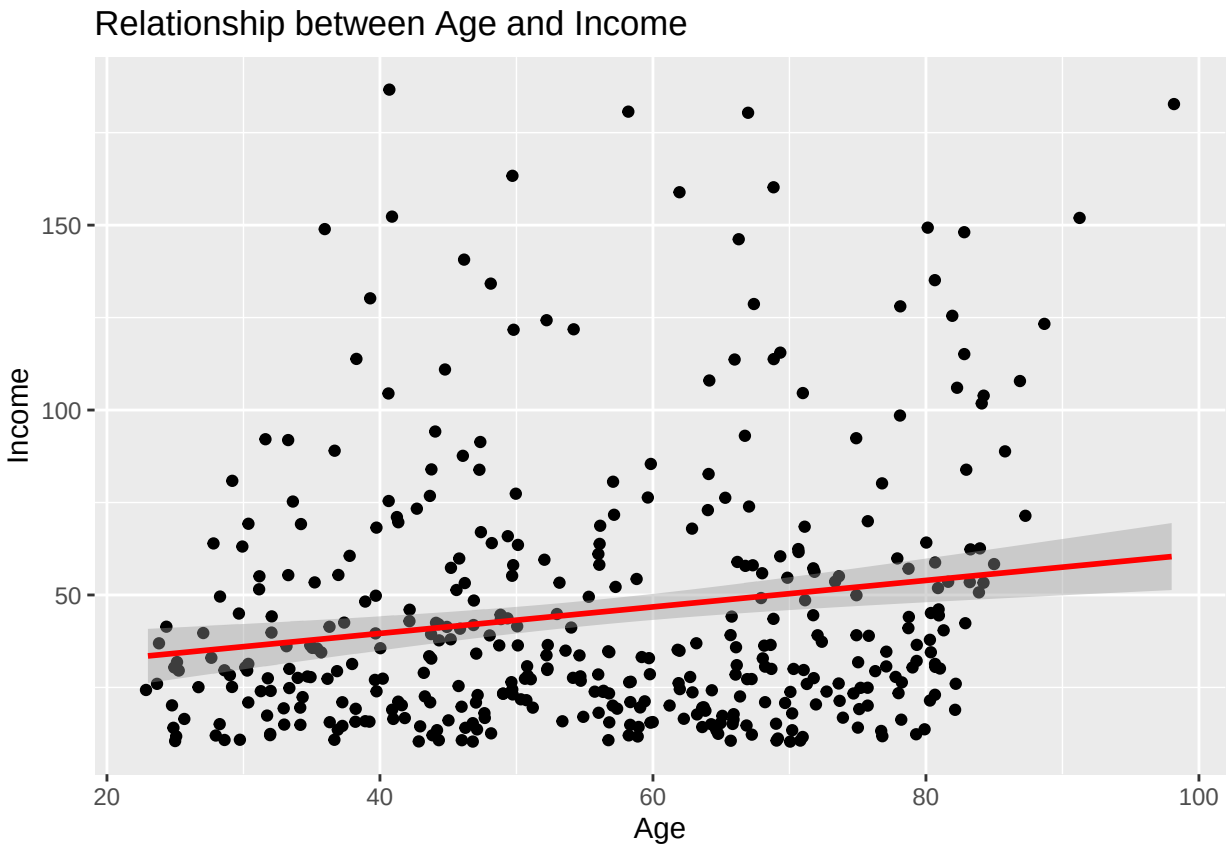
```
mod.fit <- lm(Income ~ Age, data = Credit)

mod.fit %>%
  broom::tidy()
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic   p.value
##   <chr>          <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)   25.3       5.88       4.30 0.0000213
## 2 Age            0.358     0.101      3.55 0.000426
```

```
Credit %>%
  ggplot(aes(x = Age, y = Income)) +
  geom_jitter() +
  geom_smooth(method = "lm", color = "red") +
  labs(x = "Age", y = "Income", title = "Relationship between Age and Income")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



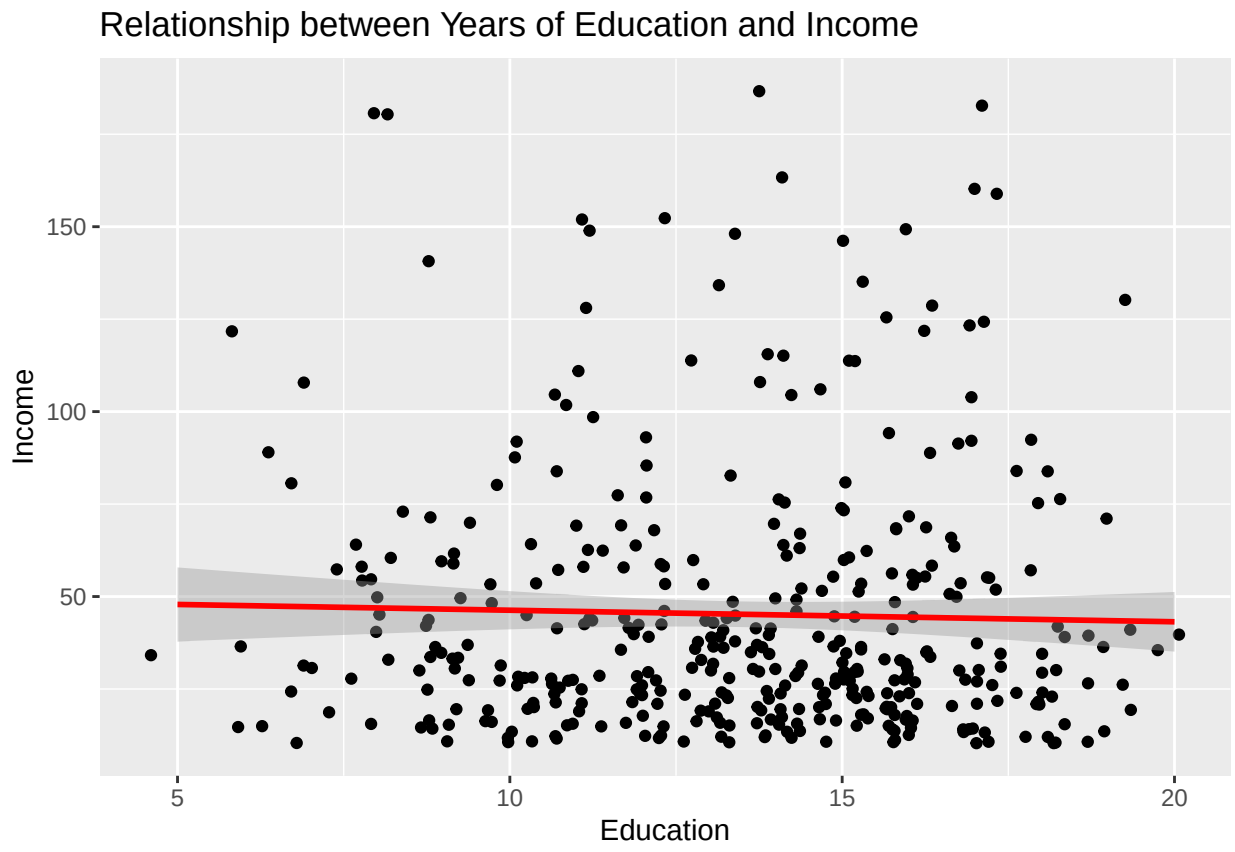

```
mod.fit <- lm(Income ~ Education, data = Credit)

mod.fit %>%
  broom::tidy()
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   49.4       7.80      6.33  6.46e-10
## 2 Education     -0.312     0.565    -0.553 5.81e- 1
```

```
Credit %>%
  ggplot(aes(x = Education, y = Income)) +
  geom_jitter() +
  geom_smooth(method = "lm", color = "red") +
  labs(x = "Education", y = "Income", title = "Relationship between Years of Education and Income")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
mod.fit <- lm(Income ~ factor(Gender), data = Credit)

mod.fit %>%
  broom::tidy()
```

```
## # A tibble: 2 x 5
##   term                 estimate std.error statistic  p.value
##   <chr>                   <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)            45.6       2.54    18.0   3.08e-53
## 2 factor(Gender)Female   -0.756     3.53    -0.214 8.30e- 1
```

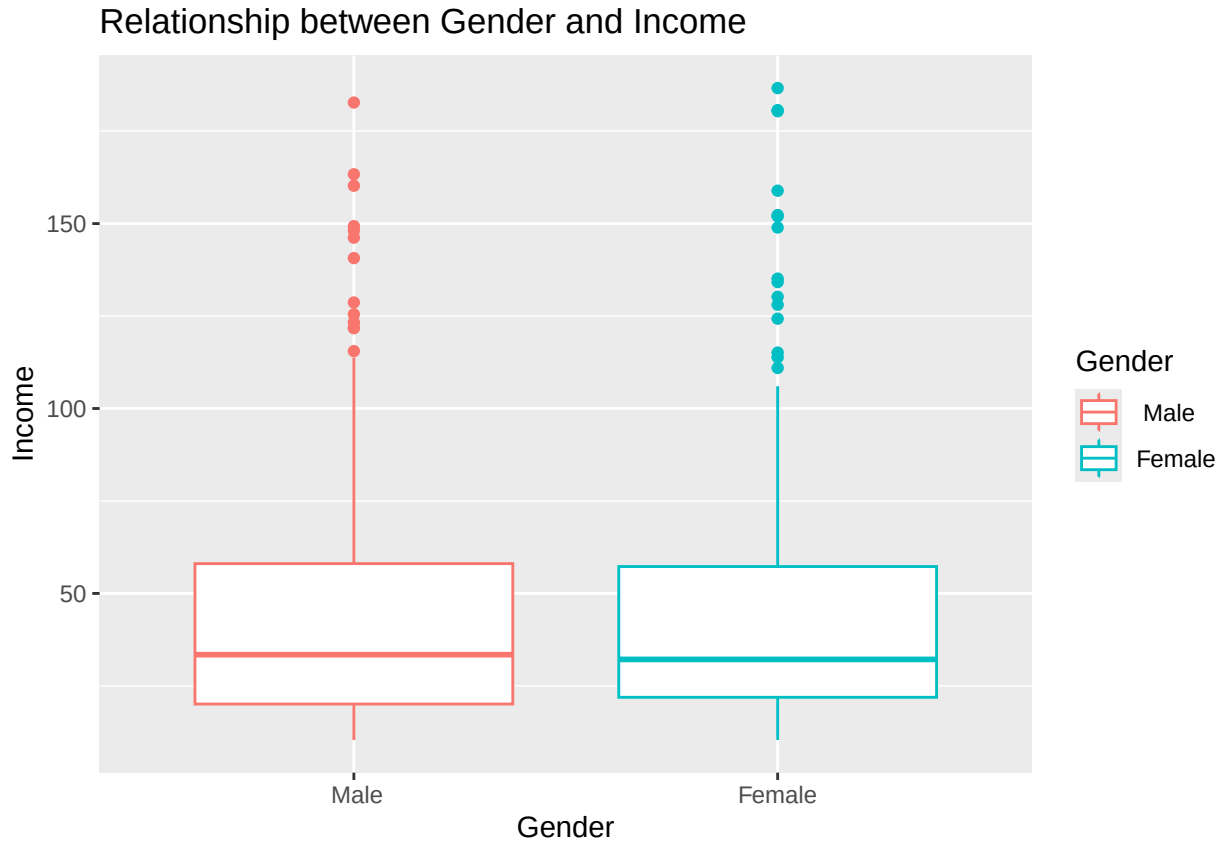
```
Credit %>% filter(Gender == "Female") %>% summarise(mean_Income_fem = mean(Income, na.rm = TRUE))
```

```
##   mean_Income_fem
## 1        44.85393
```

```
Credit %>% filter(Gender == " Male") %>% summarise(mean_Income_male = mean(Income, na.rm = TRUE))
```

```
##   mean_Income_male
## 1         45.61032
```

```
Credit %>%
  ggplot(aes(x = Gender, y = Income, color = Gender)) +
  geom_boxplot() +
  labs(x = "Gender", y = "Income", title = "Relationship between Gender and Income")
```




```
mod.fit <- lm(Income ~ factor(Student), data = Credit)

mod.fit %>%
  broom::tidy()
```

```
## # A tibble: 2 x 5
##   term                estimate std.error statistic  p.value
##   <chr>                  <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)            45.0       1.86    24.2   3.49e-80
## 2 factor(Student)Yes      2.30      5.88     0.392 6.95e- 1
```

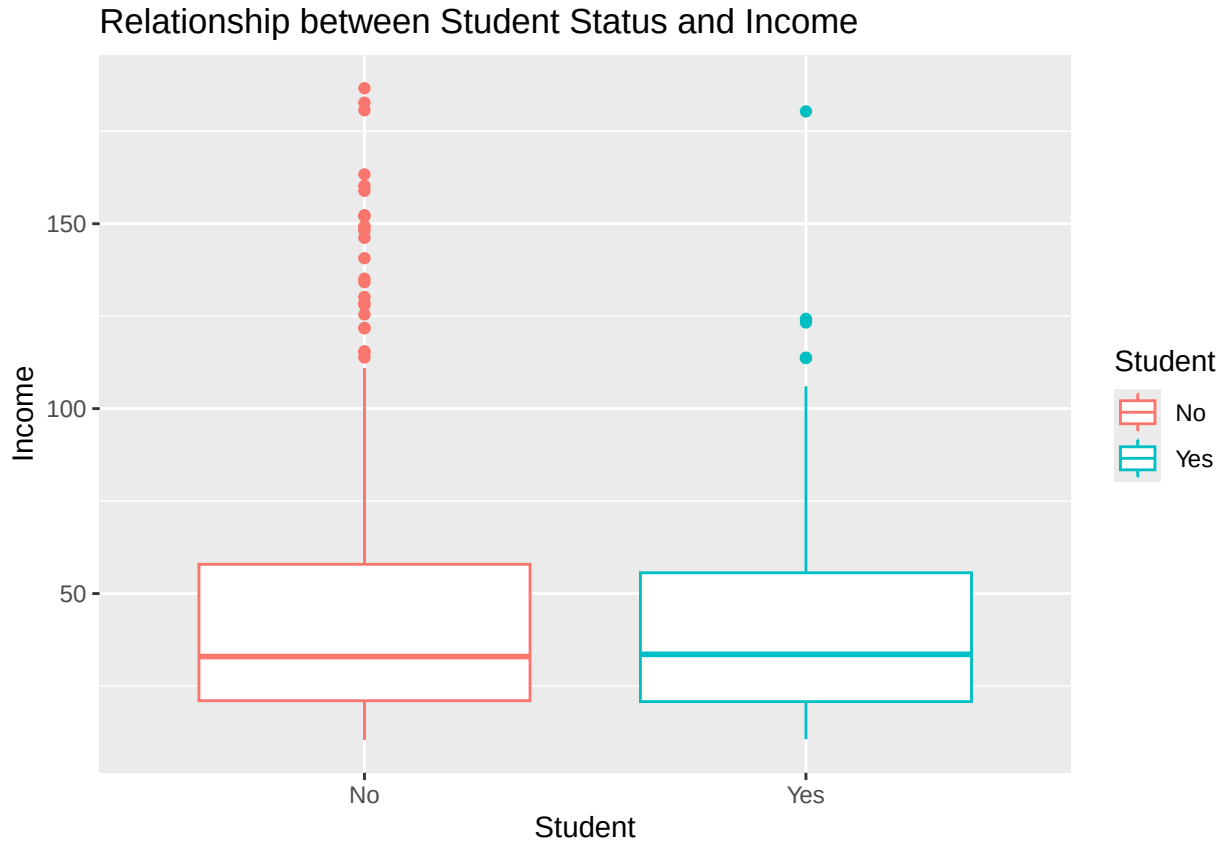
```
Credit %>% filter(Student == "Yes") %>% summarise(mean_Income_student = mean(Income, na.rm = TRUE))
```

```
##   mean_Income_student
## 1            47.29205
```

```
Credit %>% filter(Student == "No") %>% summarise(mean_Income_nonstud = mean(Income, na.rm = TRUE))
```

```
##   mean_Income_nonstud
## 1            44.98853
```

```
Credit %>%
  ggplot(aes(x = Student, y = Income, color = Student)) +
  geom_boxplot() +
  labs(x = "Student", y = "Income", title = "Relationship between Student Status and Income")
```




```
mod.fit <- lm(Income ~ factor(Married), data = Credit)

mod.fit %>%
  broom::tidy()
```

```
## # A tibble: 2 x 5
##   term               estimate std.error statistic  p.value
##   <chr>                 <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)           43.6       2.83    15.4   2.45e-42
## 2 factor(Married)Yes     2.58      3.62     0.712 4.77e- 1
```

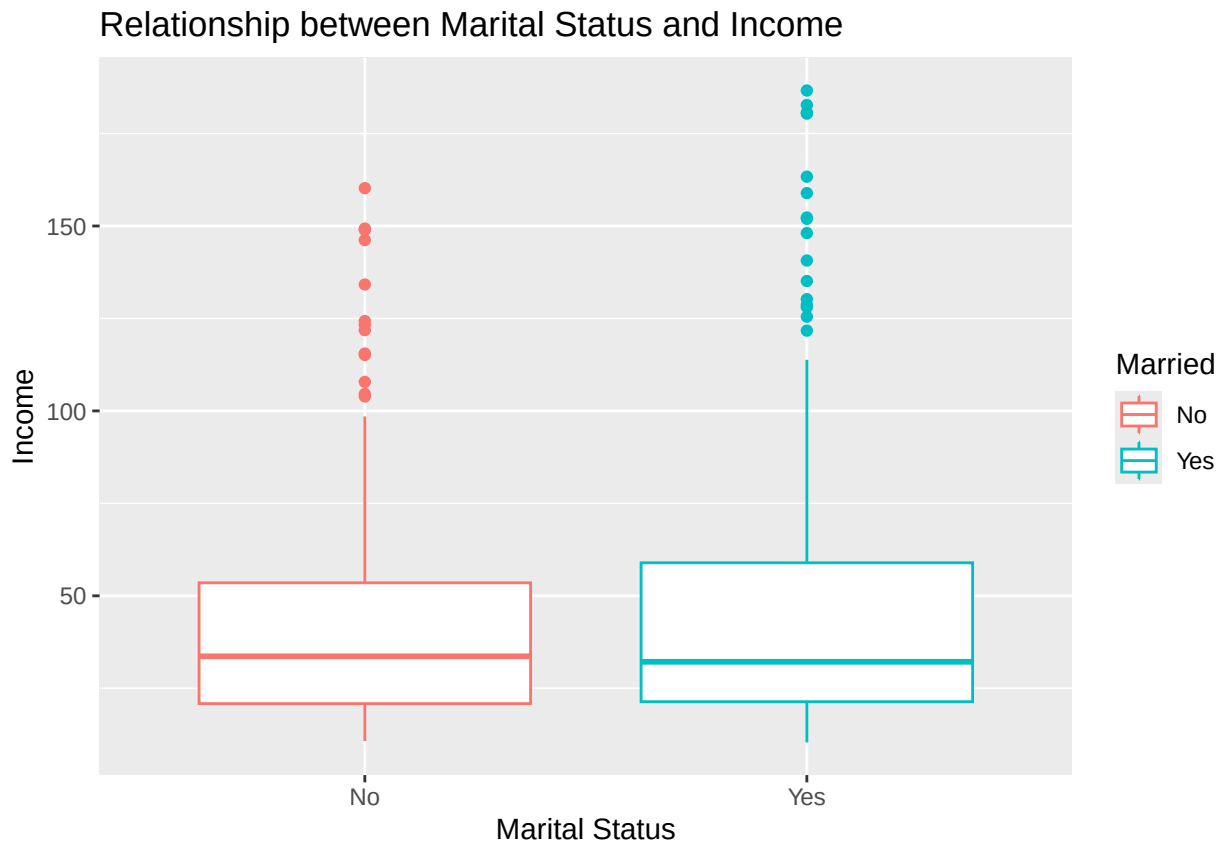
```
Credit %>% filter(Married == "Yes") %>% summarise(mean_Income_married = mean(Income, na.rm = TRUE))
```

```
##   mean_Income_married
## 1            46.21708
```

```
Credit %>% filter(Married == "No") %>% summarise(mean_Income_unmarried = mean(Income, na.rm = TRUE))
```

```
##   mean_Income_unmarried
## 1              43.64109
```

```
Credit %>%
  ggplot(aes(x = Married, y = Income, color = Married)) +
  geom_boxplot() +
  labs(x = "Marital Status", y = "Income", title = "Relationship between Marital Status and Income")
```




```
mod.fit <- lm(Income ~ factor(Ethnicity), data = Credit)

mod.fit %>%
  broom::tidy()
```

```
## # A tibble: 3 x 5
##   term                      estimate std.error statistic  p.value
##   <chr>                        <dbl>     <dbl>     <dbl>    <dbl>
```

```
## 1 (Intercept)                  47.7       3.55    13.4   3.39e-34
## 2 factor(Ethnicity)Asian       -3.49      4.98    -0.702 4.83e- 1
## 3 factor(Ethnicity)Caucasian   -3.16      4.34    -0.728 4.67e- 1
```

```
Credit %>% filter(Ethnicity == "African American") %>% summarise(mean_Income_aa = mean(Income, na.rm = T
```

```
##   mean_Income_aa
## 1        47.6821
```

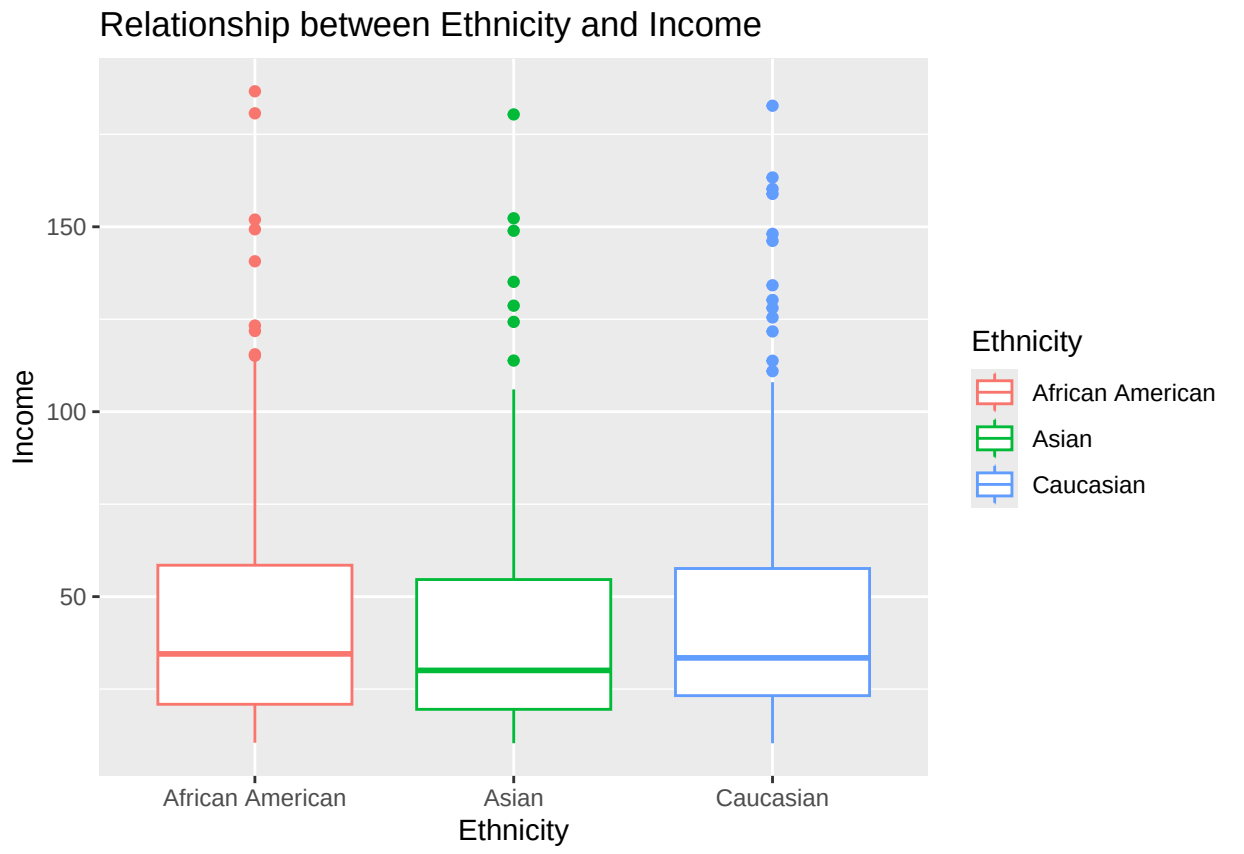
```
Credit %>% filter(Ethnicity == "Asian") %>% summarise(mean_Income_a = mean(Income, na.rm = TRUE))
```

```
##   mean_Income_a
## 1      44.18783
```

```
Credit %>% filter(Ethnicity == "Caucasian") %>% summarise(mean_Income_c = mean(Income, na.rm = TRUE))
```

```
##   mean_Income_c
## 1      44.52194
```

```
Credit %>%
  ggplot(aes(x = Ethnicity, y = Income, color = Ethnicity)) +
  geom_boxplot() +
  labs(x = "Ethnicity", y = "Income", title = "Relationship between Ethnicity and Income")
```



```
mod.fit <- lm(Income ~ Age + Education + factor(Gender) + factor(Student) + factor(Married) + factor(Et

mod.fit %>%
  broom::tidy()
```

```
## # A tibble: 8 x 5
##   term                        estimate std.error statistic  p.value
##   <chr>                          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)                   29.8      10.5       2.84  0.00469
## 2 Age                            0.365     0.102     3.58  0.000387
## 3 Education                     -0.383     0.562    -0.683 0.495
## 4 factor(Gender)Female          -0.959     3.50     -0.274 0.784
## 5 factor(Student)Yes             3.81      5.87      0.648 0.517
## 6 factor(Married)Yes             4.09      3.63      1.13  0.261
## 7 factor(Ethnicity)Asian        -2.88      4.97     -0.579 0.563
## 8 factor(Ethnicity)Caucasian    -2.87      4.31     -0.666 0.506
```

```
glance(mod.fit) %>%
  select(r.squared, adj.r.squared)
```

```
## # A tibble: 1 x 2
##   r.squared adj.r.squared
##       <dbl>         <dbl>
## 1    0.0365        0.0192
```

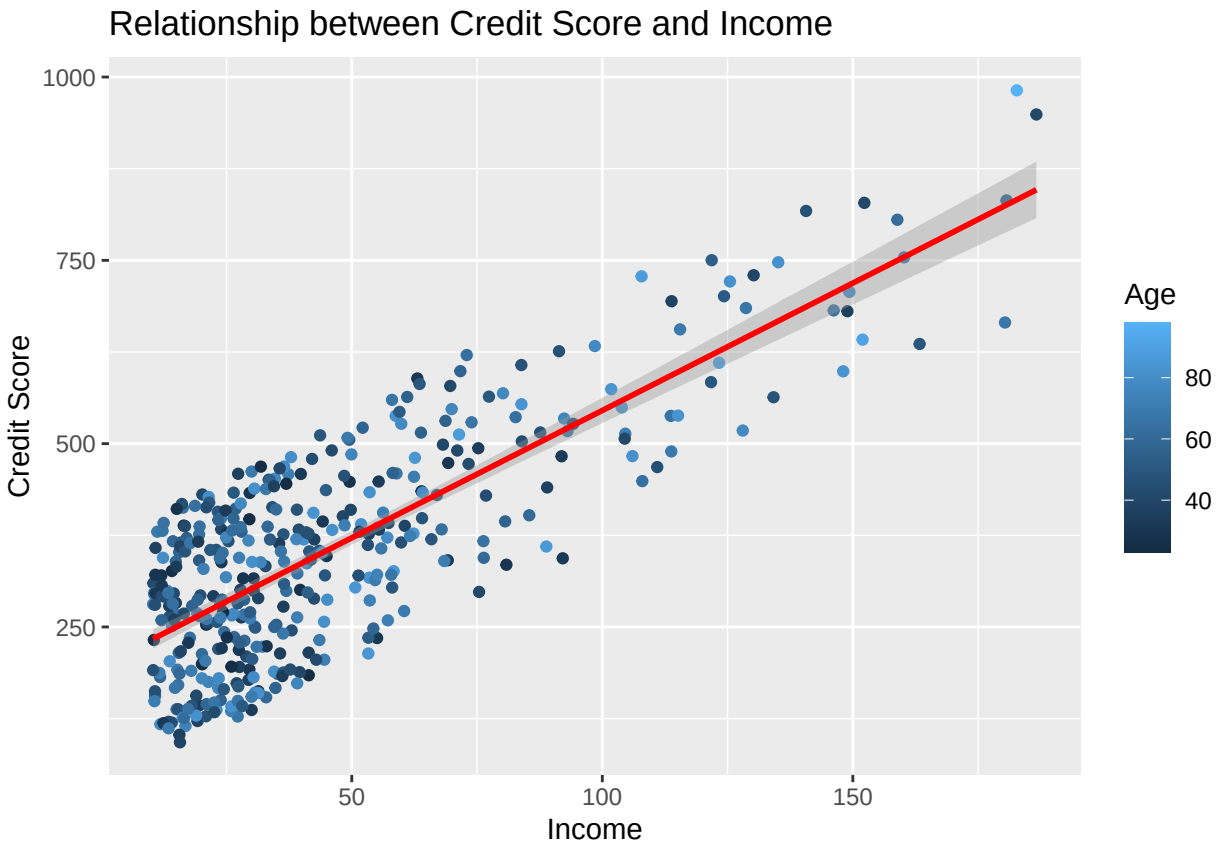
```
mod.fit <- lm(Rating ~ Income, data = Credit)

mod.fit %>%
  broom::tidy()
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   198.       7.71       25.7 2.08e-86
## 2 Income          3.47     0.135      25.8 4.36e-87
```

```
Credit %>%
  ggplot(aes(x = Income, y = Rating, color = Age)) +
  geom_jitter() +
  geom_smooth(method = "lm", color = "red") +
  labs(x = "Income", y = "Credit Score", title = "Relationship between Credit Score and Income")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```


Relationship between Credit Score and Income
Credit Score
1000
750
500
250
50
100
150
Income
Age
80
60
40